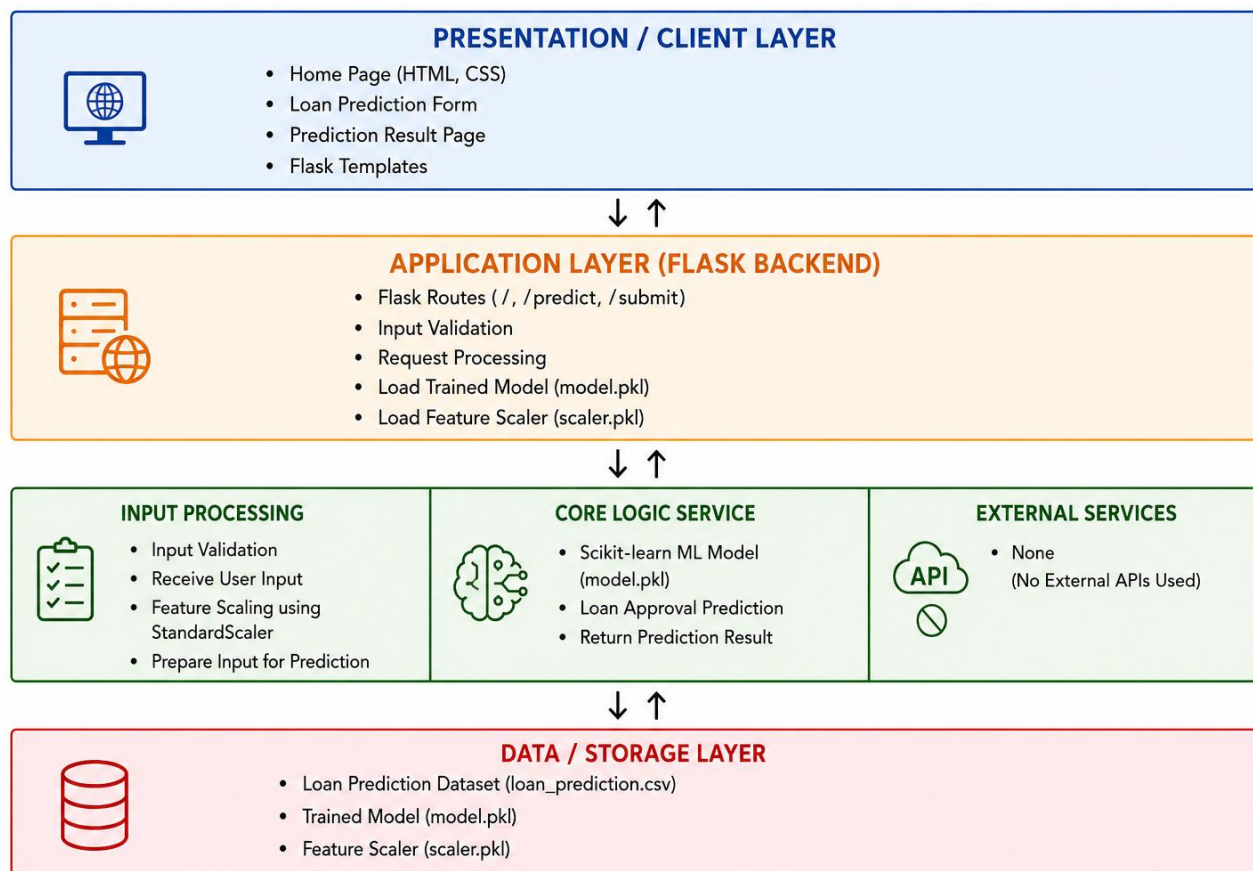


Solution Architecture

Date	27 June 2026
Team ID	XXXXXX
Project Name	Smart Lender– Loan Approval Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:

The Smart Lender solution follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Data Layer. The Presentation Layer provides a user-friendly web interface for entering loan applicant details. The Flask backend processes user requests, validates the input, scales the features using the saved scaler, and communicates with the trained Machine Learning model to generate loan approval predictions.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects applicant details and displays loan prediction results through web pages.	HTML, CSS, Flask Templates
Application Layer (Flask Backend)	Handles routing, input validation, request processing, and communicates with the Machine Learning model.	Python, Flask
Core Logic Service	Scales applicant data using the saved StandardScaler and predicts loan approval using the trained Machine Learning model.	Scikit-learn, Pickle
Data / Storage Layer	Stores the loan dataset, trained Machine Learning model, and feature scaler required for prediction	CSV Dataset, model.pkl, scaler.pkl